



auditProfessional Consulting Services for a Ground Source Heat Pump System Feasibility Study – The City of Winnipeg

Proposal for:

The City of Winnipeg
Municipal Accommodations Division
Assets and Project Management Department
185 King Street, 4th Floor
Winnipeg MB, R3B 1J1
ATTN: Lou Chubenko, Project Officer 2

Proposal Created by:

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Proposal Date:

2023 – 12 –15

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Section A: Form A – Proposal

FORM A: PROPOSAL
(See B7)

1. Contract Title
PROFESSIONAL CONSULTING SERVICES FOR A GROUND SOURCE HEAT PUMP SYSTEM FEASIBILITY STUDY – THE CITY OF WINNIPEG
2. Proponent
GEOptimize Inc.
Name of Proponent (Legal Name)

Usual Business Name of Proponent as it appears on Invoice (if different from above)
167 Lombard Avenue, unit 500
Street
Winnipeg, MB R3B 0V3
City Province Postal Code
ed@geoptimize.ca
Email Address of Proponent

Facsimile Number

(Mailing address if different)
Street or P.O. Box

City Province Postal Code

GST Registration Number (if applicable)

(Choose one)
The Proponent is:
☐ a sole proprietor
☐ a partnership
☒ a corporation
carrying on business under the above name.
3. Contact Person
The Proponent hereby authorizes the following contact person to represent the Proponent for purposes of the Proposal.
Ed Lohrenz, vice president
Contact Person Title
204-318-2156
Telephone Number Facsimile Number
4. Offer
The Proponent hereby offers to perform the Services in accordance with the Contract for the Fees, in Canadian funds, set out in the Proposal Submission.

auditProfessional Consulting Services for a Ground Source Heat Pump System Feasibility Study –
The City of Winnipeg

The City of Winnipeg
RFP No. 960-2023

Form A: Proposal

5. Commencement of the Services The Proponent agrees that no Services shall commence until it is in receipt of a notice of award from the City authorizing the commencement of the Services.
6. Contract By submitting a bid in response to this RFP, the Proponent certifies that it has read, understands, and agrees to the terms and conditions of this RFP and that the RFP, in its entirety shall be deemed to be incorporated in and to form a part of this offer notwithstanding that not all parts thereof are necessarily attached to or accompany this Proposal.
7. Addenda The Proponent certifies that the following Addenda have been received and agrees that they shall be deemed to form a part of the Contract:
- | No. | | Dated | |
|-----|-------|-------|-------|
| | _____ | | _____ |
| | _____ | | _____ |
| | _____ | | _____ |
8. Time This offer shall be open for acceptance, binding and irrevocable for a period of sixty (60) Calendar Days following the Submission Deadline.
9. Indigenous Self-Declaration The City is requesting that Proponents identify if their business is at least 51% owned by one or more Indigenous persons of Canada.
- ☐ YES, 51% or more Indigenous ownership
- ☒ NO, it is not
- This information is being gathered for statistical purposes only and will not be used for purposes of evaluation.
10. Signatures The Proponent or the Proponent's authorized official or officials have signed this
- 5th day of December, 20 23.

Signature of Proponent or
Proponent's Authorized Official or Officials



(Print here name and official capacity of individual whose signature appears above)

Ed Lohrenz

vice president

(Print here name and official capacity of individual whose signature appears above)

Section B

Fees

Below is a breakdown of the cost associated with the scope of work for each key task of the project. The scope of work is defined on page 14 and the key personnel are described in page 9.

Feasibility Study	Cost per service
Determine the ideal site based off the 12 selected sites to consider.	\$2,800.00
Conduct site visit to obtain information and review the HVAC systems on all the buildings included in the study.	\$3,600.00
Obtain other information for the buildings on site, such as, historical energy bills, building drawings, and existing building assessment studies.	\$1,800.00
Create hourly heating and cooling loads for all the buildings included in the study with representative energy models using Trane Trace 700. All the available information obtained from the site visits and provided information will be used to inform the energy models.	\$13,100.00
Determine any efficiency measures that can be applied to the buildings that will impact the GSHP system and reduce overall energy consumption.	\$3,600.00
Determine the changes required at each buildings HVAC system to be able to integrate into the GSHP system.	\$1,900.00
Determine the optimal configuration (central plant or ETP system) and size of GHX using GLD and TRNSYS. This is calculated based on the loads determined for all the buildings, as well as the soil conditions and land area available for the GHX. Vertical and horizontal GHXs will be considered in the analysis to determine the most cost effective GHX that can be installed.	\$10,000.00
Calculate a class 5 costing analysis to determine the incremental cost of the GSHP vs the business as usual. Determine the annual energy costs for the GSHP and business-as-usual case. Use current and future utility rates for gas and electricity to determine the NPV and lifecycle cost analysis of the GSHP system.	\$3,700.00
Determine the CO2 emissions savings based on carbon emission intensities for producing electricity and burning natural gas, and the calculated energy consumption for the GSHP system and business as usual case.	\$2,300.00
Determine the type of monitoring and maintenance required for the GSHP system.	\$2,000.00
Develop a schedule for the installation of the proposed GSHP system that includes the entire construction process from beginning to completion to commissioning.	\$3,800.00
Develop a list of incentives that might be available for the project, including local, provincial, and federal incentives.	\$2,000.00
Develop and finalize report summarizing all the details of the feasibility study.	\$17,950.00
Cost	\$68,550.00
GST	\$3,427.50
Total (Inc. GST)	\$71,977.50

Pricing Summary

Description	Line Total
GSHP Feasibility Study	\$68,550.00
GST	\$3,427.50
Total (including 5% GST)	\$71,977.50

Section C

About GEOptimize

The use of fossil fuels to heat and cool buildings is being restricted around the world. Building and vehicle electrification is forcing more efficient use of electricity. Ground source heat pump (GSHP) systems are powered with electricity to deliver heating and cooling needs more efficiently than any other available technology.

GEOptimize focuses on optimizing the cost and performance of GSHP systems in homes, commercial, and industrial buildings. GEOptimize also focuses on large scale district geothermal grids that take

advantage of waste energy from refrigeration systems, wastewater, combined heat and power, and renewable energy resources.

Conventional heating and cooling systems are connected to a gas or electric utility. Designers are typically not concerned about the amount of energy needed during a year, but only with the instantaneous peak heating and cooling loads.

A GSHP system is connected to a ground heat exchanger that takes advantage of the energy storage capacity of the earth. To determine how it will perform, a designer needs to know:

- The amount of energy the ground heat exchanger must deal with instantaneously & annually.
- Thermal properties of the earth the GHX is built in.
- The land area available to construct the GHX.
- The capabilities of the contractors in the area that can construct the ground heat exchanger.

GEOptimize uses commercially available software to model building performance and works closely with the project design team to optimize the building and mechanical systems to perform well with a GHX for the life of the building. For larger systems with multiple buildings and other energy resources, GEOptimize has developed customized modules using TRNSYS to simulate the performance of district systems such as university campuses and community energy sharing systems. GEOptimize is certified to provide professional engineering services in the province of Manitoba.

GEOptimize has developed an extensive library of hourly building energy models of many building types. These are modified to simulate the performance of your proposed building project with a GSHP system and compare it to other types of systems from an economic, technical, and environmental perspective to quickly assess their feasibility.

When constructed, GEOptimize works with you and your building operator to develop an understanding of how the system is designed to perform and to continue optimizing system performance into the future. Ongoing monitoring of your system is essential to allow for changes in the building, how it is used and operated and compensate for changes in the weather from year to year. GEOptimize strongly recommends that all GSHP systems are continuously monitored to ensure optimum performance. Relevant project experience is in Appendix C.

GEOptimize experience with GSHP systems and ice rink integration

Ed Lohrenz has worked in many facets of the GSHP industry since 1982, including work as a contractor, with a manufacturer, and since 1995 as a consultant in the geothermal heat pump industry. Since 1989 Ed has been instrumental in the design and implementation of over 50 ice rink and ice storage systems across Canada and the U.S.

Background information about GEOptimize

GEOptimize was founded by Ed Lohrenz in 2013. Since then, the company has grown to include 3 mechanical engineers and drafting personnel. GEOptimize focuses on the design and implementation of ground source heat pump systems. The typical scope of work completed by GEOptimize includes:

- Detailed hourly energy modeling of the project buildings to determine the energy loads the ground heat exchanger (GHX) must deal with. Typically, several iterations of an energy model are completed in an effort to balance the energy transfer to and from the ground as this results

in the smallest, least costly GHX, and the GHX that performs most reliably for the life of the building.

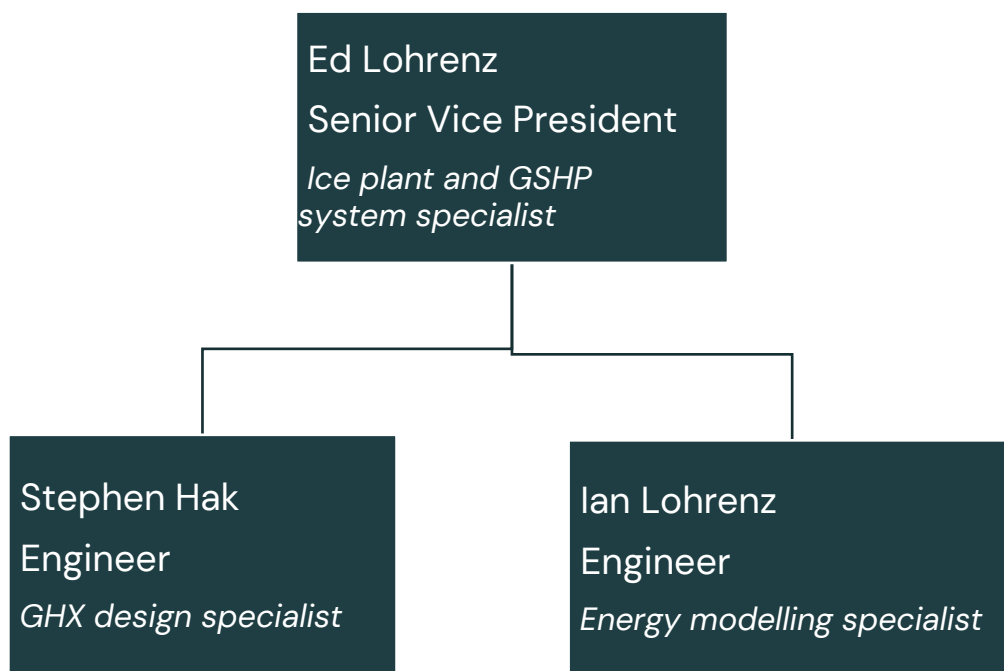
- Ground heat exchanger design based on the geology of the site, the land area available to build it, and the ability of the contractors in the area that can construct it. Determining and testing the geology of the site is an important component of the design of a cost-effective GHX.

Many of the clients of the firm include mechanical engineering firms that have not had the opportunity to become involved in the design of the ground heat exchanger. GEOptimize works closely with the mechanical designers of the building mechanical system to ensure the system is compatible with the GHX the system is connected to.

In December 2022, GEOptimize was acquired by Endurant Energy LLC. Endurant Energy offers engineering, procurement, and construction (EPC) services to clients across the U.S. and Canada. Endurant also offers energy as a service (EaaS) contracts to clients across North America.

Section D

The feasibility study will be conducted by three key personnel for this project. Ian Lohrenz's primary role will be creating the energy models for the project in Trane Trace 700. Stephen Hak will design the GHX using GLD (Ground Loop Design) and TRNSYS. Ed Lohrenz will mainly be in a supervisory role providing his experience. Each key personnel will help develop the final report. Below is an organizational chart for this project.



Key Personnel

Ed Lohrenz, BES, CGD



Ed has worked in the ground source heat pump industry since the early 1980's. He has been fortunate to have worked with many people as passionate about the technology as he is. He has had the opportunity to design GSHP systems for many applications and in climates ranging from the Australian outback to the Siberian tundra as well as many unique projects across Canada and the U.S.

Comparable Projects Completed by Ed Lohrenz

- Packer Arena, Austin, MN: The 30,000 ft² community ice arena in Austin, MN was constructed in 2002. Waste heat from the refrigeration plant provides heating for the arena, the arena seating, domestic hot water for showers and ice resurfacing and the snow melt system.
 - Facility manager: Steve Moe – P: 507 437-7704
- South Cariboo Recreation Centre, 100 Mile House, BC: The 32,000 ft² community ice arena was constructed in 2003. Waste heat from the refrigeration plant provides heating for the arena, the arena seating, domestic hot water for showers and ice resurfacing and the snow melt system.
 - Facility manager: Josh Dickerson – P: 250-395-1353 E: jdickerson@canlansports.com
- Dr. Carson and Marion Murray Community Centre, Springhill, NS: The 31,000 ft² community centre was constructed in 2004. Waste heat from the refrigeration plant provides heating through a radiant floor heating system, seat heating, domestic hot water for showers and ice resurfacing. A supply and return water well system absorbs excess energy from the refrigeration plant when heat is not required and is a heat source for the arena when heat is not available from the refrigeration system.

Facility manager: Cory Skinner – P: 902-667-2313 E: cskinner@cumberlandcounty.ns.ca

Stephen Hak, EIT, BSc



Stephen is a Biosystems Engineering graduate from the University of Manitoba. He is currently pursuing a Master of Science in Engineering with a focus in geothermal energy. He has gained significant experience using TRNSYS for simulating the performance of ambient temperature, multi-building district geothermal systems, and experience in energy modelling, ground heat exchanger design, and ground source heat pump systems. Stephen has completed over 10 district GSHP feasibility studies.

Comparable Project – Community of Inverness Nova Scotia

This project was a feasibility study focused on converting several buildings within the community of Inverness, Nova Scotia into a GSHP system. The buildings that were included in the study are: hospital, k-12 school, nursing home, Grocery store and local hockey rink. Several other sources of energy were

analysed including wastewater heat extraction, open well heat exchanger, and heat source/sink from a local abandoned coal mine well. For this project we looked at maximizing the waste heat from the arena to provide enough heating for the buildings within the study. The feasibility study determined the economic and environmental impact of the GSHP based off the results of the energy models and conceptual design of the district system. Stephen's main role was to design the conceptual district GSHP system and complete the report describing all the details of the feasibility study. The owner of this project was the Municipality of the County of Inverness who hired Eastpoint Engineering as the mechanical engineer. Eastpoint Engineering hired GEOptimize as the subconsultant.

Contact 1: Peter Simpson, Eastpoint Engineering, 1-902-456-7763, peter.simpson@eastpoint.ca

Contact 2: Kathy Hannigan, Municipality of the County of Inverness, 1-902-258-7535,

Kathy.hannigan@invernesscounty.ca

Comparable Project – University of Minnesota Morris

The goal of this project was to conduct a feasibility study which focused on converting 28 buildings on the University of Minnesota Morris campus from a central boiler chiller system to a district GSHP system. Looked at incorporating open well heat exchangers with horizontal directionally drilled ground heat exchangers. Stephen's role was to design the district GSHP system, conduct the economic analysis, determine the carbon emissions savings, and produce the final report. Ottertail Power Company hired GEOptimize to conduct the feasibility study for the University of Minnesota Morris.

Contact 1: Jon Fabre, Ottertail Power Company, 1-218-739-8633, jfabre@otpc.com

Contact 2: Brian Herrmann, Vice Chancellor University of Minnesota Morris, 1-320-589-6113

herrmanb@morris.umn.edu



Ian Lohrenz, P.Eng, CGD

Ian has focused on energy efficiency in buildings since graduating from the University of Manitoba with a Bachelor of Science in mechanical engineering in 2008. Through experience and education, he has developed an expertise in building energy modelling, ground heat exchanger design, ground source heat pump systems, building commissioning, and energy auditing.

Comparable Project – Missoula Triangle DES Study

GEOptimize studied the potential for utilizing GSHPs in a district energy system at the Montana Rail Link (MRL) Triangle site in Missoula, MT in 2023. The impact of rooftop solar was included in the analysis. Ian created the detailed energy models used by the team to size the GSHP system and the PV analysis. The Municipality of Missoula hired ICF, and ICF hired GEOptimize to provide our expertise on the GSHP system.

Contact 1: Craig Schultz, ICF Senior Advisor, 1-202-669-6200, craig.schultz@icf.com

Contact 2: Maggie Irwin, ICF Environmental Specialist, 1-703-251-0778, Maggie.Irwin@icf.com,

Comparable Project – Bethel Place Deep Retrofit Study

In 2023 Ian worked with a local team including SEEFAR Building Analytics, Bird Construction, Number 10 Architects, Epp Siepman Engineering, and Crosier Kilgour, to study the feasibility of several deep energy retrofits at this seniors' apartment block owned by Bethel Mennonite Care Services, in Winnipeg including converting the existing boiler system to a GSHP system. Ian conducted the ASHRAE Level 2 Energy Audit, created a calibrated model, and proposed energy model used to size the GHX and determine energy savings of the project.

Contact 1: Jim Nolsted, SEEFAR Building Analytics, 431-373-8255, info@seefar-valuation.com

Contact 2: Richard Marshall, Bird Construction, 204-784-3186, Richard.Marshall@bird.ca

Section E

Project Understanding

The City of Winnipeg is looking to determine the technical and economic feasibility of installing a GSHP system at one of the arenas located in Winnipeg. The goal will be to maximize the waste heat from the arena to incorporate into the GSHP system to reduce natural gas consumption, resulting in fewer CO₂ emissions to meet Winnipeg's Climate Action Plan and the Community Energy Investment Roadmap. The GSHP system will provide heating and cooling for the arena and any buildings nearby to reduce the overall carbon emissions and energy consumption by replacing their current HVAC systems with a GSHP system.

The project will involve choosing an optimal site for the GSHP system based on the information provided for 12 potential sites. Once the site is determined, a feasibility study will be conducted to convert the buildings on site to a GSHP system that utilizes the waste heat from the arena.

Methodology

Site Selection

The first step of the project will be to identify the best location for the feasibility study based on 12 locations provided in Table 1 Appendix A from the RFP. The choice for the most suitable project will be based off the following criteria:

- Land area available for the GHX.
- The current condition of the HVAC system at the arena including the ice plant.
- The number of buildings near the arena that can be incorporated into the system to maximize energy savings.
- If any heating loads are required nearby, such as a pool.

The location will be determined to maximize energy savings while minimizing the capital and operational costs. This section will be completed by all the key personnel for this project.

Energy Modelling

To quantify the individual buildings space heating and cooling loads accurately with limited data, previously constructed building energy models (using Trane Trace 700 or Trane Trace 3D) from an energy model database using Winnipeg, Manitoba weather data will be used to extract peak heating and cooling loads and total heating and cooling energy loads for the space usages. The available information obtained for the buildings such as historical energy bills, building drawings, and existing building assessment studies will be used to inform alterations to the energy models.

Several iterations of the energy models will be created and used for the analysis. Based on the information provided, various efficiency measure upgrades will be considered based on the current building conditions. Potential efficiency measure upgrades that will be considered are upgraded lighting, upgraded envelope u-factors, upgraded windows, or adding an energy recovery ventilation (ERV) system. These efficiency measures will be applied to manipulate and ultimately reduce the heating and cooling loads to ensure an optimized GSHP system.

The information obtained will also be used to determine the technical feasibility of converting the buildings to a GSHP system from their current HVAC systems.

Ian Lohrenz will complete the work for this section of the study.

GHX Modelling

The performance of a GHX is dependent on:

- Total annual heat transfer to and from the building(s)
- Peak instantaneous heat transfer to and from the building(s)
- Ambient earth temperature at the location
- Thermal characteristics of the earth in which the GHX is constructed
- Detailed design and configuration of the GHX

It is recommended at this stage in the project, that no thermal conductivity (TC) test be conducted. A TC test can cost up to \$15,000 and should only be conducted after it is determined that the project is feasible. If the thermal conductivity test is completed, and the feasibility study indicates that the project should not proceed, then the money used for the TC test is wasted. The calculated energy loads based off the representative models will have a much larger impact the size of GHX required compared to the thermal properties of the soil. Thermal properties and temperature of the soil can be estimated with reasonable accuracy in Winnipeg, as the geological conditions are consistent. The soil consists of clay up to 40–60 ft depth and the remaining soil is fractured limestone. If no geotechnical information is available for this site, GEOptimize will use information obtained from previous jobs in Winnipeg to estimate the thermal properties of the soil at the site.

The land area available for construction of a GHX will be determined with feedback from the City of Winnipeg and any relevant stakeholders at the project site. The dimensions and layout of the land designated for constructing GHX can influence the required depth of vertical boreholes to fulfil the project's heating and cooling demands. Alternatively, these factors may also indicate the suitability of horizontal GHXs for the site.

To ensure the longevity and proper system operation, it is highly recommended that the GSHP be monitored in terms of the amount of energy flowing in and out of the GHX.

GEOptimize uses GLD software to simulate the performance of vertical and horizontal GHX's. Both vertical and horizontal GHXs will be considered in this study. Refer to Appendix B for more technical details relevant to horizontal and vertical GHXs.

Stephen Hak will complete the work for this section of the study.

District System Modelling

Different types of buildings have distinctly different hourly energy load profiles. A school, for example, is typically occupied during a weekday, unoccupied on the weekends and evenings and almost always unoccupied in summer. A residential building is occupied heavily from late afternoon till the following morning, and less heavily occupied during a weekday, while more heavily occupied on a weekend. Peak heating and cooling loads in the buildings do not coincide.

Connecting buildings with different energy load profiles to a system that can transfer energy from one building to another will be beneficial in terms of system efficiency and GHX size. Soil and rock are poor heat conductors. Heat absorbed by the earth surrounding the GHX piping takes time to dissipate. The energy is stored in the soil around the pipe until it dissipates or is drawn from the ground when heat is needed in the building. The ground stores excess heat when it's not required in other buildings and releases it when it's not available from other buildings. For this project, a district system will maximize the waste heat generated from the ice plant if other buildings are connected to the system, especially if these buildings require more heating than cooling on an annual basis.

GLD can only model GHXs for a single load set (or one building). To account for the energy loads of all the buildings, the compressor heat from the ice plant, and the compressor heat from all the heat pumps at each building, TRNSYS software is used. TRNSYS is a software that GEOptimize uses to model district GSHP systems. The modelling will determine the optimal GSHP system, in terms of the configuration (central plant or Energy Transfer Pipe (ETP) system), of the district GSHP system and the amount of GHX required for the overall system. Appendix A describes the difference between a central plant system, and the ETP system.

Stephen Hak will complete the work for this section of the study.

Economic & Emissions Analysis

Class 5 cost estimates will be developed for the conceptual design proposed. This will include budget costing for the supply and install of the GHX, distribution infrastructure and any recommended efficiency measures on a cost per square foot basis. Energy consumption of the existing systems will be compared with the estimated energy consumption of the proposed conceptual GSHP system. Using the energy consumption and the current and future rates for natural gas and electricity, the Net Present Value (NPV) and lifecycle cost savings of the GSHP will be determined to compare to the business-as-usual case.

CO₂ emissions of the current heating and cooling systems will be compared with CO₂ emissions of the proposed GSHP heating and cooling system. CO₂ intensity (CO₂/kWh) of electricity production from Manitoba Hydro and from burning natural gas will be used to calculate emissions for the business-as-usual case and the GSHP system.

This section will be worked on by all the key personnel for this project.

Scope of Work

- Determine the ideal site based off the 12 selected sites to consider.
- Conduct site visit to obtain information and review the HVAC systems on all the buildings included in the study.
- Obtain other information for the buildings on site, such as, historical energy bills, building drawings, and existing building assessment studies.
- Create hourly heating and cooling loads for all the buildings included in the study with representative energy models using Trane Trace 700. All the available information obtained from the site visits and provided information will be used to develop the energy models.
- Determine any efficiency measures that can be applied to the buildings that will impact the GSHP system and reduce overall energy consumption.
- Determine the changes required for each buildings HVAC system to be able to integrate into the GSHP system.
- Determine land area available for the GHX, estimate soil conditions based off any geotechnical information available for the site or from previous experience.
 - It is not recommended to conduct TC test until the project is deemed feasible.
- Determine the optimal configuration (central plant or ETP system) and size of GHX using GLD and TRNSYS. This is calculated based on the loads determined for all the buildings, as well as the soil conditions and land area available for the GHX. Vertical and horizontal GHXs will be considered in the analysis to determine the most cost effective GHX that can be installed.
- Calculate a class 5 costing analysis to determine the incremental cost of the GSHP vs the business as usual. Determine the annual energy costs for the GSHP and business-as-usual case. Use current and future utility rates for gas and electricity to determine the NPV and lifecycle cost analysis of the GSHP system.
- Determine the CO₂ emissions savings based on carbon emission intensity for producing electricity and burning natural gas, and the calculated energy consumption for the GHSP system and business as usual case.
- Determine the type of monitoring and maintenance required for the GSHP system.
- Develop a schedule for the installation of the proposed GSHP system that includes the entire construction process from beginning to completion to commissioning.
- Develop a list of incentives that are available for the project, including local, provincial, and federal incentives.

Deliverable

GEOptimize will deliver a comprehensive written report on the GSHP feasibility study, encompassing all elements outlined in the work scope. This report will equip the City of Winnipeg with a thorough comprehension of the methodologies and procedures undertaken to obtain the study's findings.

Consequently, the city will be able to apply the insights gained from this report to future projects of similar nature.

Section F

Project Schedule

A project schedule was determined based off the scope of work. The Gantt chart below estimates the start date to be January 1st, 2024. It was calculated that it will take approximately 11 weeks to complete the scope of work for the feasibility study.

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Appendix A: District System Distribution Configurations

District systems have evolved over time since the earliest systems were built in the late 1800's (Figure 1). The most significant change has been a reduction in temperatures of the heat transfer fluid delivered to buildings connected to it. As delivery temperatures have dropped the systems can take advantage of low-grade waste heat sources (heat recovery from building cooling & refrigeration systems, wastewater, etc.) and renewable energy sources has increased overall system efficiency. Figure 1 illustrates the steady increase in system efficiency as the temperature of the heat transfer fluid to connected buildings has decreased since the late 1800's.

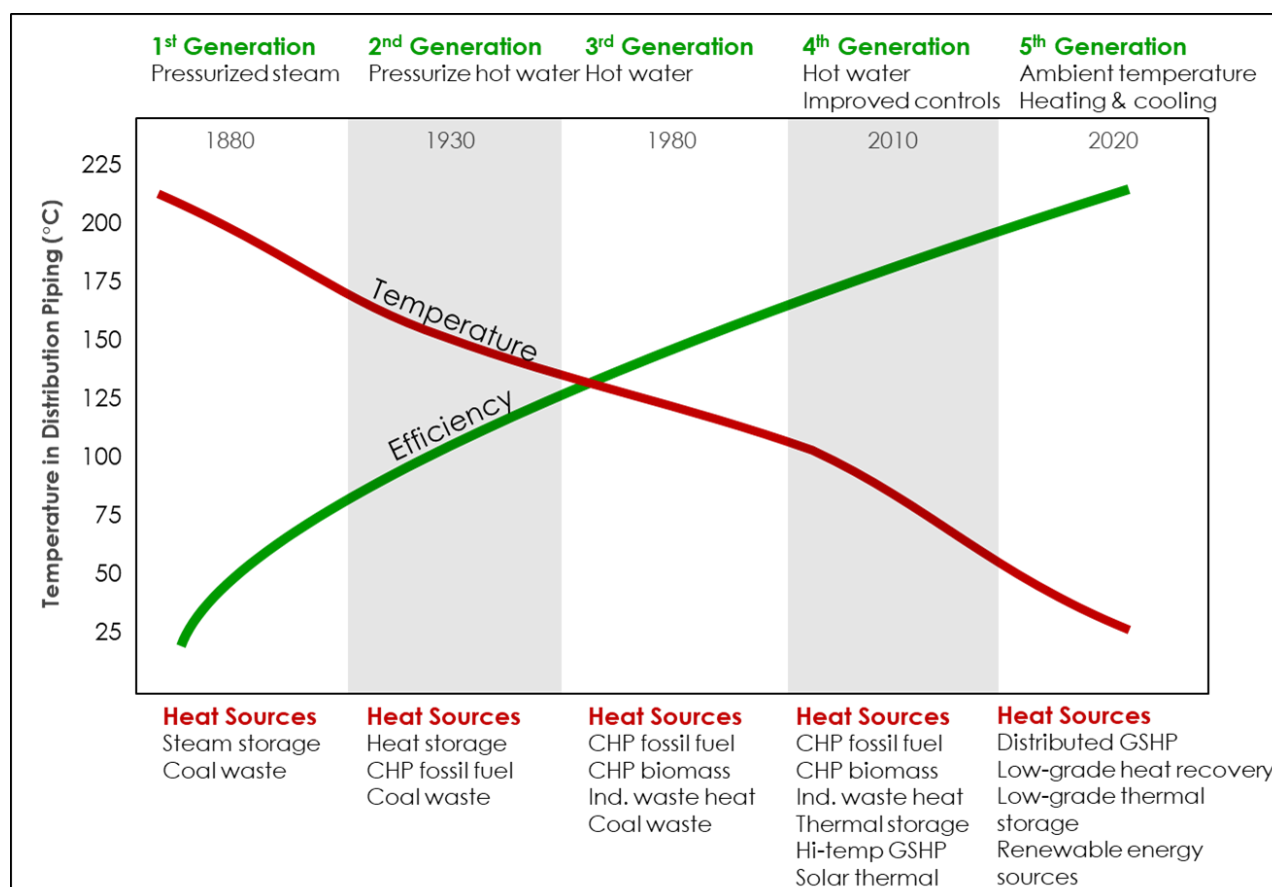


Figure 1: History of district energy systems and temperature vs. efficiency

4th generation district energy systems (DES's) are designed with a central energy plant that produces hot and chilled water. Two pairs of insulated supply and return pipes circulate hot and chilled water from a central energy plant to each building on the DES. If a building requires heat, hot water is pumped from the hot water supply line to the building distribution system. The temperature of the hot water is cooled as the building is heated and is pumped back to the central energy plant from the return line. Similarly, when a building is cooled, heat is rejected from the building to the chilled water and returned to the central energy plant through the return pipe (Figure 2).

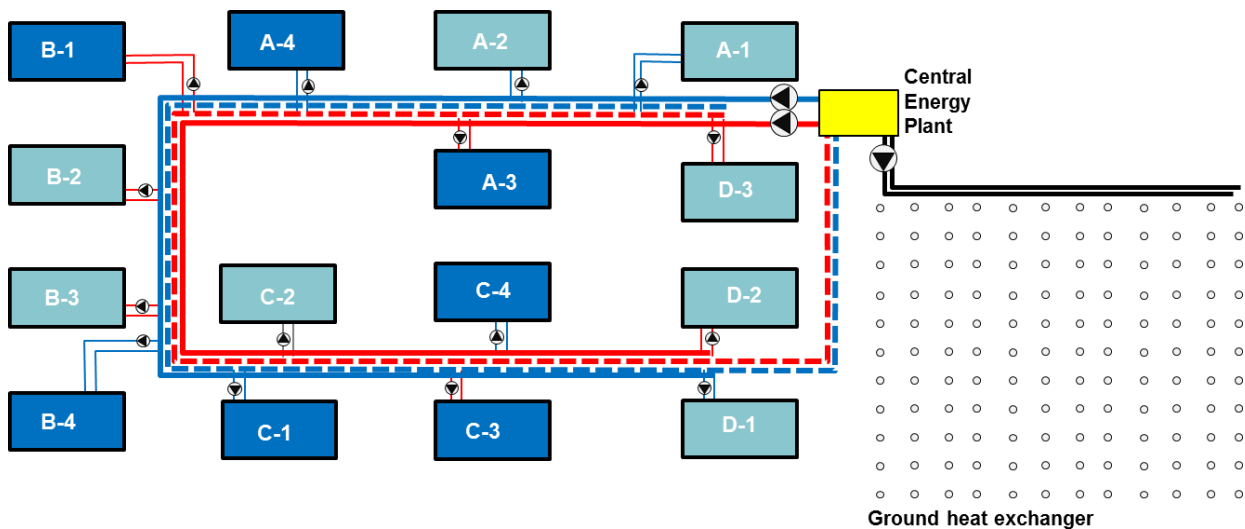


Figure 2: 4G district GSHP energy system

The main disadvantages of a 4th generation District Energy Systems (DES's) are:

- Distribution piping and circulation pumps must have the capacity to meet the combined peak heating and cooling loads of all buildings connected to the DES.
- DES delivers same water temperature to all buildings. The building with the highest water temperature governs the supply water temperature of the entire system reducing overall system efficiency.
- Requires full planning of future system to ensure central plant has capacity to meet heating and cooling demands of completed development although development construction may not be completed for several years.

5G DES's takes advantage of energy sharing by moving energy from a building that is cooling and rejecting heat to the system to another building that is extracting heat from the system. When the amount of heat rejected to the energy transfer pipe (ETP) is equal to the amount of heat extracted from the ETP, energy does not need to be added to or removed from the system. When the temperature of the ETP falls outside normal operating parameters (4°C/40°F to 29°C/85°F) the energy source is activated to maintain the normal operating temperatures. Rather than using a boiler as the energy source or cooling tower to dissipate heat, a pump connected to a ground heat exchanger (GHX) is activated to add or remove energy from the ETP (Figure 3).

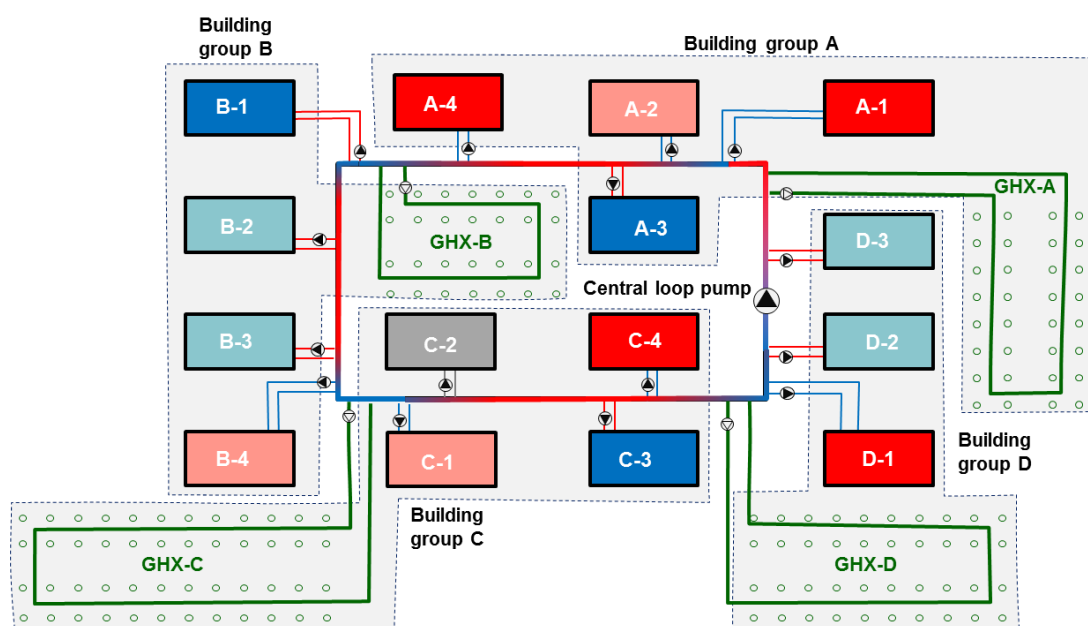


Figure 3: 5G district GSHP energy system

The main advantages of 5G DES's are:

- A single ETP operates at GHX temperature and does not require insulation as delivery temperature to buildings is not critical, reducing initial capital cost.
- Mechanical systems in each building are designed and optimized for the distribution system in the building without compromising the efficiency of the entire system. Buildings can be connected to the ETP at any location without affecting system operation which allows the system to adapt to changes as the needs of the client change.

Appendix B: Vertical & Horizontal Closed Loop Ground Heat Exchangers

Vertical GHX's consist of a plastic HDPE U-bend that can be installed into the ground to a depth up to 1,000 ft. The benefits to a vertical GHX are if you have restricted land area, you can increase your heat exchange surface area by drilling deeper vertically rather than using land horizontally.

Horizontal directionally drilled (HDD) GHX's consist of drilling a pilot hole with a directional drilling rig and then pulling a U-bend back through the pilot hole (Figure 4). The drill rig can install piping as deep as 30 ft and can have 3 sets of u-tubes stacked vertically to a depth of 30 ft, 20 ft, and 10 ft, for example. There are several advantages of an HDD GHX. These include:

- Lower capital costs
- Minimal site disturbance (horizontal boreholes can be installed under existing paved parking areas or even under buildings)
- The proximity of the horizontal GHX piping to the atmosphere allows dissipation of excess heat to the atmosphere in winter and warm outdoor air temperatures and rainfall can add energy to the GHX during the summer, reducing the potential for long-term temperature degradation.

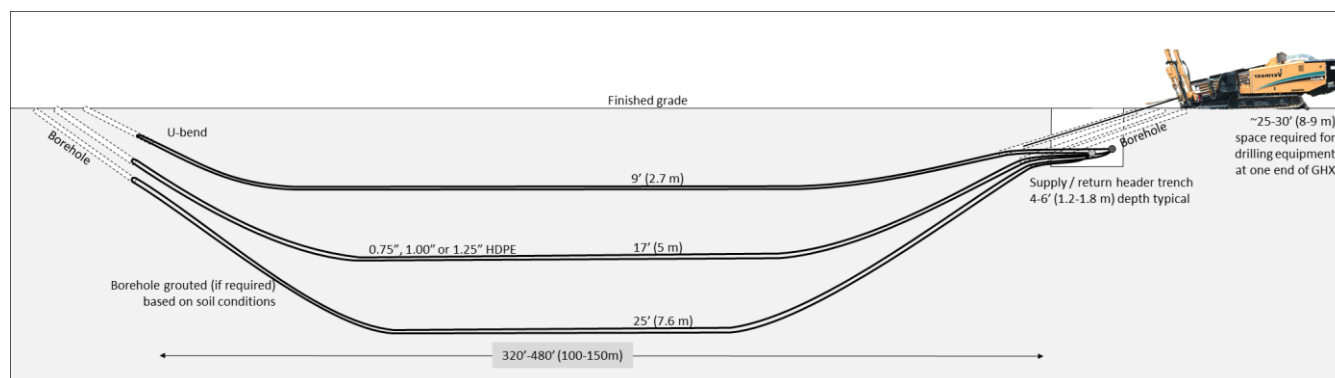


Figure 4: Visual representation of horizontal directionally drilled boreholes.

Appendix C: Previous GSHP Feasibility Studies

MRL Triangle in Missoula, MT

With support from U.S. EPA Office of Brownfields and Land Revitalization (OBLR), ICF and GEOptimize conducted a clean energy analysis for the City of Missoula, Montana. The clean energy analysis focused on the potential for utilizing ground source heat pumps (GSHPs) in a district energy system at the Montana Rail Link (MRL) Triangle site. The impact of rooftop solar was included in the analysis. MRL Triangle is an 11.42 acre former industrial and commercial site that Missoula is considering for redevelopment.

The team evaluated four different energy scenarios:

1. Conventional dual fuel

- Space heating and domestic hot water (DHW): Natural gas
- Space cooling: Electricity

2. Conventional all-electric

- Space heating and DHW: Electrical unit heaters and baseboards and standard electric DHW tanks
- Space cooling: Electricity

3. High-performance all-electric using air source heat pump (ASHP)

- Space heating and DHW: ASHP using electricity and standard electrical DHW tanks.
- Space cooling: ASHP using electricity.

4. High-performance all-electric using GSHP

- Space heating and DHW: GSHP derived electricity
- Space cooling: GSHP derived electricity

A conceptualized mixed-use site redevelopment was modelled consisting of 14 buildings with commercial, retail, and residential space for a total of 386,538 ft² of conditioned space. Building energy modeling, ground heat exchanger (GHX) modeling, district GHX modeling, and solar PV modeling were conducted to quantify the energy consumption of each scenario and to design the

solar PV and the district GSHP system. The analysis revealed that GSHPs are the most efficient way to heat and cool a building. The GSHP scenario costs more to install, but the installation costs are mitigated by lower energy demand, energy cost savings, and incentive programs. Using a District 5G energy sharing system known as an energy transfer pipe (ETP) that contains a district pipe loop with 72 vertical bores at 400 ft depth, and 3 groundwater wells with a capacity of 40 gallon per minute (gpm) could maximize the district energy system's efficiency and cost-effectiveness. The incremental capital cost of installing the GSHP system in comparison to the other scenarios was estimated at \$1,375,849. The solar modeling revealed that implementing solar energy would reduce annual energy consumption by 55% on average across all buildings.

Carbon emissions from the MRL Triangle buildings are estimated at 707 metric tons (MT) of CO₂e in the High-Performance All-Electric scenario (Scenario # 4) before GSHP and solar are added. GSHP and solar then reduce carbon emissions by a combined total of 70%, with 34% attributable to GSHP and 36% to solar.

An important note about this report is that the solar and GSHP systems are physically and economically independent. The estimated net effects of solar, which are reported separately in this report, can also be deducted from any of the scenarios to evaluate solar impacts alone, including energy use, carbon reduction impacts, and economic impacts.

*Reference: Craig Schultz, Senior Advisor, Renewable Energy, ICF, 1-202-669-6200,
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Bemidji State University

Bemidji State University (BSU) is a public state university located in Bemidji, Minnesota, USA with over 5,400 students and 23 buildings on campus with 1,450,000 ft² of floor area. BSU embarked on a plan to achieve carbon neutrality by 2050. GEOptimize was retained to determine the feasibility of installing a district GSHP system in the facility and compared it to a business-as-usual case with replacing the current central plant natural gas heating and electric cooling system.

To quantify the individual buildings space heating and cooling loads, previously constructed building energy models using Trane Trace 700 were used. Based on the various buildings subdivided space usage, hourly energy loads from the existing energy models were scaled either up or down depending on the floor area difference and aggregated to create a new building energy profile. This method was applied to each of the 23 buildings in this analysis. Individual energy load profiles were then optimized by using energy efficiency measures of lighting upgrades and energy recovery ventilators (ERV's). These efficiency measures allowed for a reduction in overall space heating and cooling while balancing the energy loads. This resulted in a reduced capital cost for GHX and reduced energy consumption.

The energy load profiles were used to model the size and performance of the GHX using Ground Loop Design (GLD) 2016 software and thermal properties from the soil profile on campus. A horizontal directionally drilled (HDD) GHX was determined to be the most cost effective due to abundant land area on the campus and relatively low-cost drilling rates. Based on various interviews with the BSU staff and information sharing, a building priority list was developed to portion the capital cost required to begin the project. A modular construction process was applied by utilizing energy transfer pipe technology.

Based on the methodology described, in depth financial calculations were performed for each construction step based on the GHX cost, distribution piping cost, incentives, and energy cost savings. Two in depth analyses were performed, one with efficiency measures applied to the buildings and one without. The analysis without efficiency measures was deemed unfeasible due to a 40% greater GHX cost. Additionally, the impact from carbon dioxide (CO₂) emissions comparing a GSHP and conventional heating and cooling system was performed.

Reference: Roger Garton, Senior Commercial and Industrial Representative, Ottertail Power Company, 1-218-739-8542, rgarton@otpc.com.

Wallen, Australia Housing Development

This project looked at the feasibility of installing a district geothermal system at a new housing development in Wallan, Australia, which is 60 km north of Melbourne, Australia. The development consisted of 7,000 single family homes with two schools, a social centre, and a hotel. This project looked at incorporating a district geothermal system for 290 single family homes.

Due to the limited space on each individual lot for a GHX, the design consisted of one vertical GHX in each lot with a large GHX located in the parks within the development. The single vertical GHX at each lot was calculated to provide approximately 60% of the total capacity of the home. By connecting the homes to a common district geothermal system, the remaining capacity could be provided by the large GHX fields located at the parks in the development. To model the district geothermal system, TRNSYS software was used.

This study found when comparing a conventional natural gas furnace heating and electric A/C cooling system to a district geothermal system, the electrical consumption increased by 50% while reducing the electrical demand by 29%. This showed that the geothermal system has a great benefit to the electric utility by increasing their sales without having to increase their electricity production. The study also found that on average, the annual utility bills of each home would be reduced by approximately 56% when comparing to a natural gas furnace heating and electric A/C cooling system.

Reference: Max Ploumis, Engineering Manager, Direct Energy Australia, +61 3 9546 8666, max@directenergy.com.au

University of Minnesota, Morris

The University of Minnesota, Morris (UMM) is one of five universities in the University of Minnesota system. The campus currently has 39 buildings totaling 970,000 ft² of floor area situated on 100 acres of land on the eastern side of Morris, Minnesota. This analysis included 28 of the buildings on campus connected to the central heating and cooling plant. UMM has embarked on a plan to achieve carbon neutrality by 2025. GEOptimize was retained to determine the feasibility of installing a district GSHP system on the campus and compare it to the current central plant natural gas heating and electric cooling system.

To design a GSHP system, hourly heating and cooling loads of the campus were estimated based on building size and use. The loads were calibrated to the current energy consumption on campus. Individual energy load profiles were then optimized by integrating energy efficiency measures of lighting upgrades and energy recovery ventilators (ERV's). ERV's transfer energy from exhaust air to

pre-heat or pre-cool incoming outdoor ventilation air to reduce the heating and cooling load in the buildings. More efficient lighting and lighting controls reduced the cooling load as less energy is added to the space. These efficiency measures allow for a reduction in overall space heating and cooling while balancing the energy loads. This resulted in a reduced capital cost for ground heat exchangers (GHX) and reduced energy consumption.

An energy transfer pipe (ETP) system moves energy from one building that is rejecting heat to the system (cooling) to another building that is extracting energy from the system (heating). It can be thought of as a conveyor belt that moves energy around the system. A ETP district style GSHP system was analysed with and without efficiency measure. The analysis without efficiency measures was deemed unfeasible due to a 33% greater GHX cost. UMM currently plans on incorporating an energy storage system for the electricity produced on campus. The effects on energy cost and CO₂ emissions from the anticipated energy storage system was analysed.

The total incremental cost for the district GSHP system including efficiency measure upgrades was \$8,446,342 with annual energy cost savings of \$310,886. The capital cost did not include the mechanical distribution systems, as this cost would be similar regardless of the heating and cooling system type. Since the construction process is split into steps, this allows for compounding energy savings as more buildings are connected to the ETP.

The current energy consumption at UMM is 1,178,516 therms and 8,188,518 kWh annually. Converting to a GSHP with efficiency measures, the campus energy demand will be 11,932,823 kWh of electricity and 27,895 therms of natural gas. The remaining natural gas used for heating buildings was not included in the analysis. UMM can currently generate on average 10,303,521 kWh annually from solar and wind. If this energy is stored, the annual energy bill at UMM will be \$93,978 which is 85% lower than what is currently paid annually. UMM emissions after the retrofit are projected to be 1,098 MT of CO₂ which is 87% lower than the current emissions at UMM. Incorporating the GSHP system enables UMM to use all the electricity produced on campus.

*Reference: Jon Fabre, Supervisor Marketing Programs, Ottertail Power Company, 1-218-739-8633,
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